Info Problemset V

aperson31415

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Various problems which don't need much observation, mostly for training common algorithms. Sorry if the typesetting is a little weird, I migrated most of these over from markdown. trust trust practice these problems btw. These are a lot harder than the problems I usually make though

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Powers of n

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are investigating the last few digits of powers of numbers.

Input Format

The first line of input contains 3 space separated integers n, m, and k

Output Format

Output the last k digits of n^m

Subtasks & Constraints

For all subtasks, it is guaranteed that $n, m, k \geq 0$, and $n, m \leq 10^5$

Subtask	Points	Additional Constraints
1	5	$n \in \{-1, 0, 1, 2, 5, 10\}$, $n^m \leq 10^18$
2	10	$n \in \{-1, 0, 1, 2, 5, 10\}$
3	10	$k \leq 1$
4	20	$k \leq 5$
5	25	$k \leq 10^3$
6	30	$k \leq 10^5$

MNAS

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob knows of the classic DP MNASS problem, but he isn't experienced enough to understand the solution. That's why he is first tackling the less common MNAS problem (Minimum Null Acceptable Square). In this problem, you are given a grid of integers, and basically you need to find the maximum sized square containing only 0s in it.

Input Format

The first line of input contains 2 integers, r and c, the number of rows and columns in the grid. The next r lines of input each contain c space separated integers, the values of each grid point.

Output Format

Output a single integer, the side length of the MNAS of the given grid.

Subtasks & Constraints

For all test cases, $0 \leq r, c, \leq 10^3$ and the number on each grid cell is between -10^9 and 10^9

Goodbye to the NZIC(s)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are holding a farewell party for the NZIC (New Zappy Integrated Circuit) after having been deemed *too unsafe* for play use. With the last supply of N NZICs, you want to get as much enjoyment as you can. Each NZIC has a score s_i, meaning it can zap up to s_i people before becoming useless.

You have N friends, who you will each give 1 NZIC to. Then, they will play the classic laser tag game, shooting at each other. In this game, each friend is in either team 1 or team 2. Given M actions where friend i shoots player j which increases the team that friend i is in by 1 (obviously only if the NZIC is still useful), find the maximum sum of the value of team 1 score + team 2 score.

Input Format

The first line of input contains 2 integers N and M. The second line of input contains N integers, the scores of each NZIC. The third line of input contains N integers, the team of each friend, and the fourth line of input contains M lines of input, each with 2 integers i and j.

Output Format

Output a single integer, the maximum sum of the 2 teams' scores.

Subtasks & Constraints

For all subtasks, $N, M \geq 0$, and the number of friends in each team will be around the same.

Subtask	Points	Additional Constraints
1	5	$N, M \leq 10$
2	5	$N, M \leq 20$
3	10	$N, M \leq 100$
4	30	$N, M \leq 1000$
5	50	$N, M \leq 10^6$

Pie a la Mode (Redux)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

The local patisserie sells the most delicious pastries. Your favourite kind is the pie a la mode, a fluffy fruit-filled pie served with a scoop of specialty ice cream. There are many different flavours of icecream to be had, and you never know which one you are going to get.

Lately you have been using your laptop to keep track of which icecream flavours you have recieved. Of course, typing in the name of each flavour is tedious and error-prone, so instead you have used numbers to represent different flavours. (For example,0=vanilla,1=chocolate,2=raspberry sorbet...)

However, your sister has also been using your program, but you know she **always** buys the same ice-cream as you. As you have a grudge against her, you don't want your favourite ice-cream to be the same as her's, so you want the program to output your second favourite ice-cream.

Wondering which flavour you have gotten the most on each day, you set to work writing a program.

Input Format

The first line of input contains 1 integer n, the number of days you recieve icecream. The next n lines of input each contain 1 integer, the type of icecream you recieved on each day.

Output Format

Output n lines of input, each with 1 integer, your second most bought type of ice-cream over days 1 to i.

Subtasks & Constraints

For all test cases, $1 \leq n \leq 10^6$ and there are no more than 10^9 flavours of icecream.

Welcome to the AIO!

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You were bored, so you searched on github, and somehow found an AIO(T) tutorial, and since you had nothing better to do, you clicked on the link and started. However, you immediately realised that it was not the AIO you meant to find, it was the All-In-One (Tool) tutorial, not the Australian Informatics Olympiad (tutorial) tutorial :(But since you always finish what you start, you must get to work and finish as fast as you can.

There are N pages, each which take a_i time to read through, and M one-way links between pages, which take b_i seconds to load. Answer Q queries of the form: if starting at page d_i, what is the minimum time to read all pages you go through, while clicking links between those pages, and reach one of the K certificate pages which are the pages labelled with $c_1, c_2, \ldots, c_K$.

Input Format

The first line of input contains 3 integers, N, M, K, and Q. The 2nd line is a, the 3rd line is b, the 4th line is c, and the 5th line is d.

Output Format

For each of the queries, output a single integer on a new line of output, the minimum time to start from page d_i and reach one of $c_1, \ldots, c_K$. If it is not possible to do so, output -1 for such queries.

Subtasks & Constraints

For every test case, $N, M, K, Q \geq 0$, and $a_i, b_i, c_i, d_i \geq 0$, and all b_i are generally significantly smaller than all a_i

Subtask	Points	Additional Constraints
1	10	$N, M, Q, K \leq 100$
2	10	$N, M, Q, K \leq 1000$
3	15	It is not possible to reach the same page after clicking a series of links
4	15	$K = 1$ and $N, M, Q \leq 10^5$
5	15	$a_i, b_i = 1$ for all i
6	35	$N, M, Q, K \leq 10^5$

Powers of n (Redux)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

After investigating the last digits of powers of numbers, you have decided to embark on a more ambitious investigation, investigating the front digits of powers of numbers!

Input Format

The first line of input contains the *base* n, the number which would be the base of the power. The next line of input is a number a, the required first few digits of a power.

Output Format

If there exists an integer k such that the first $|a|$ digits of n^k are a, then output the smallest possible k. Else, output -1.

Subtasks & Constraints

For all subtasks, it is guaranteed that $n, a \geq 1$, $n \leq 10^18$ and a has no more than 10^6 digits.

Subtask	Points	Additional Constraints
1	30	You only either need to output 1, or -1, seeing if it is possible for an integer k to exist
2	70	No additional constraints

Additioning (Interactive)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are solving the annoying problem Addition (Troll), especially the last 0.1 Mark subtask when you realise that you could just systematically spam submit different guesses. Therefore, your task is to make a *submitter* for Addition (Troll).

Input Format

As usual, your first line of input will be N.

However, the next part is that you can call the function ‘float guess(int a, int b)‘ to guess ‘a‘ and ‘b‘, giving your score for the current Addition (Troll) problem (see scoring section for how this worked). You will be marked accordingly to how many calls of this function you take to correctly guess A and B.

Subtasks & Constraints

For all subtasks $N \geq 3$, and A and B will always be positive integers and distinct.

Subtask	Points	Additional Constraints
1	10	$N \leq 4$
2	20	$N \leq 10^3$
3	70	$N \leq 10^6$

However, since this problem is kinda hard, you will be scored accordingly to the scoring function, where a is my best result, and b is yours, you will score $b/a\%$ for that test case.

For the original troll problem, you would obtain 1 mark if you had the correct sum ($A + B = N$), and 2 additional marks if A and B were the correct numbers, but swapped around.

Nomnomnomnomnom: Where am I?

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Just to be safe that no kinds were playing games, the Department of Education has added filtering to their internet services, instantly flagging any device using a VPN, emulator, playing unblocked games, or being off-task. However, results showed that in the top schools, around 90% of devices were not following these internet rules, so this surely was a bug right?

Anyway, lots of people started playing Battle Cats using BlueStacks or LDPlayer and somehow, the Super Metal Hippoes escaped into the Sydney CBD! Since humans are no match for Super Metal Hippoes, you have been invited to work with researchers at UNSW's Hippo fusion machine, allowing you to fuse different traits of different Hippoes around Sydney.

Each hippo has a health h_i and an attack a_i (both integers). When combining Hippoes, you can use a real number percentage of any combination of available hippoes to create new ones (like you could have 50.67% of hippo 1, 10pi% hippo 2, 49.33-10pi% hippo 3).

However, UNSW's hippo researchers were not really ready for this situation, so the created hippoes must have an integer health and attack.

Your task is that, given available hippoes, how many hippoes can UNSW's fusion machine create?

Input Format

The first line of input contains n, the number of currently available hippoes. The next n lines of input each contain 2 space-separated integers h_i and a_i.

Output Format

Output a single integer, the number of hippoes UNSW's fusion machine can create. Since this result might be big, output it modulo 10^9

Subtasks & Constraints

For all test cases, $1 \leq n \leq 10^6$.

Subtask	Points	Additional Constraints
1	10	$-10^4 \leq h_i, a_i \leq 10^4$
2	30	$-10^9 \leq h_i, a_i \leq 10^9$
3	60	$-10^18 \leq h_i, a_i, \leq 10^18$

Oddjobs (Redux)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

There are N interns who have practically done nothing all day, other than raiding the snacks and playing table tennis. No more! You, their manager have exactly N jobs that need doing. Thus you have decided to assign one intern to each of the jobs.

You've determined for each intern, how many mistakes they will make for each job you might assign them. Not really wanting to make more work for yourself, you've decided to assign interns to job so as to minimise the total number of mistakes made.

Input Format

The first line of input will contain the single integer $1 \leq N \leq 100$, the number of interns(and the number of jobs).

N lines follow, each containing N integers. On the ith line, the jth integer denotes the number of mistakes the ith intern will make doing the jth job.

Each intern will make somewhere between 0 and 10^6 mistakes on any given job.

Output Format

Output the fewest total mistakes you can achieve if you assign jobs to interns optimally.

Trout 0: Ancestors

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are investigating the family tree of trout in New Zealand, and want to know who was the *greatest* trout of all time. The *greatness* of a trout is the sum of all it's children, grandchildren, great-grandchildren...'s *achievement* score (of course they have to be alive, and the trout has to be alive).

You are given n events in chronological order, each where either a trout dies, or a trout is born, and your task is to find the maximum *greatness* of any trout at any time.

Input Format

The first line of input is 2 space separated integers, n and m. The second line of input contains m space separated integers, the achievement scores of each trout. The next n lines of input are as follows:

- '-1 x': Trout x dies
- '1 x y': Trout x is born, with parent y

Output Format

Output a single integer, the maximum *greatness* of any trout at any time. Since this result might be quite large, output it modulo $10^9 + 7$

Subtasks & Constraints

It is guaranteed that some obvious stuff is true, like a dead trout never makes a child, and a non-existent trout never dies. It is also guaranteed that $x_i, y_i \in \{1, m\}$ for all i.

Additionally, $n, m \leq 10^6$.

Hooliganism II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

The City of Perth has been having problems with "hooliganism" as of late. Out of control youths are driving around without permits, scaring hippoes, driving over daisies, and crashing into each other. Now you must help implement a plan to lock out these hooligans once and for all, while allowing normal citizens to move around.

The City of Perth is made up of N intersections, numbered 1 to N, connected by R roads. Each night all roads in Perth will become one-way, to stop collisions. You decide which way each road goes. To ensure that normal citizens can drive wherever they want though, every intersection should be reachable from every other intersection.

Input Format

The first line of input contains 2 space separated integers, N and R. The next R lines of input contain 2 integers a_i and b_i, showing that intersection i connects road a_i and b_i together.

Output Format

If such a configuration is not possible, output '-1'.

Else, you should output R lines of input each containing 2 space separated integers x_i and y_i showing that intersection i goes from x_i to y_i in a one-way direction.

Subtasks & Constraints

For all subtasks, $N, R, x_i, y_i \geq 0$.

Subtask	Points	Additional Constraints
1	25	You will be marked if you output '-1', or not (determine if a configuration is possible)
2	75	$N, R \leq 10^7$

Negotiations 2: High school

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

In a distant eastern corner of the high school, beyond the vast sandpits and just over the Salty Puddle, there is a small strip of playground that has been the centre of some commotion. While most of the children get along with each other and take turns with the equipment, Isobelle and Paul have been squabbling over this strip for countless lunchtimes. The feud has become so intense that many other children are now arguing about the dispute, with an increase in threats and "mysterious rickrollings" that could possibly lead to an undesirable all-out playground war. As Isobelle's chief negotiator, you are keen to bring the squabble over the disputed playground strip to an end once and for all, by separating an area for Paul to play and an area for Isobelle to play next to each other in peace.

The playground is separated into a grid of r rows by c columns of *plots*, each with a certain play value. Find the kth largest possible sum of a rectangle of grid cells.

Input Format

The first line of input contains the integers r and c. The next r lines contain c integers, the play value of each cell of the playground

Output Format

Output a single integer, the size of the kth largest sum of a rectangle of grid cells.

Subtasks & Constraints

For all test cases, $R, C > 0$ and $R \leq 100, C \leq 400$

Breaking Cars

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You and your friend would like to share breaking cars. The carpark is an grid of car parking lots, with cars positioned at x_i and y_i, each with satisfaction of breaking s_i.

You and your friend would like to split the carpark into 2 regions separated by a straight line, such that the difference between you and your friend's satisfaction is minimised.

Input Format

The first line of input is n, the number of cars in the carpark. The next n lines of input contain 3 space separated integers x_i, y_i, and s_i.

Output Format

Output the minimum possible difference between you and your friend's satisfaction. Since this result might be big, take the result modulo $10^9 + 7$

Subtasks & Constraints

For all subtasks $n \leq 10^3$, $\max(s_i) \leq 10^{12}$ (over all i)

Subtask	Points	Additional Constraints
1	40	$\max(x_i, y_i) \leq 10^9$ (over all i)
2	60	$\max(x_i, y_i) \leq 10^{15}$ (over all i)

A not so simple sort (Redux)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob is studying informatics and he has just discovered sorting algorithms! However, he is kinda bad at them, so he wants you to sort some lists for him.

You are given a list of $n \geq 0$ positive integers, with maximum value of $x \geq 0$. Output the list, sorted in ascending order.

Input Format

The first line of input contains a single integer n. The second line of input contains n space separated integers, the elements of the list.

Output Format

Output n space separated integers on a single line, the sorted list, ascending.

Subtasks & Constraints

Subtask	Points	Additional Constraints
1	2	$n \leq 9$ and $x \leq 10^{18}$
2	4	$n \leq 10$ and $x \leq 10^{18}$
3	6	$n \leq 10^3$ and $x \leq 10^{18}$
4	8	$n \leq 2 \cdot 10^5$ and $x \leq 10^{18}$
5	10	$n \leq 2 \cdot 10^6$ and $x \leq 2 \cdot 10^6$
6	15	$n \leq 2 \cdot 10^6$ and $x \leq 10^{18}$
7	20	$n \leq 2 \cdot 10^7$, $x \leq 2.5 \cdot 10^7$, and no 2 elements are equal
8	35	$n \leq 2 \cdot 10^7$, $x \leq 10^7$

It is better to solve the last few subtasks separately.

Knapsack (Hard)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob is studying more informatics, and has discovered Dynamic Programming, specifically the knapsack problem. He knows the $O(N^2)$ and $O(NW)$ solutions, but this problem is too hard for him! Please help him solve it.

You are given n items, with weight w_i and score s_i. Find the maximum amount score you can gain by choosing items such that their weight sums to x. Additionally, you can only choose at most k of each item.

Input Format

The first line of input contains 3 space separated integers, n, k, and x. The next n lines of input contain 2 space separated integers w_i and s_i.

Output Format

Output a single integer, the maximum score you can gain. However, since this result might be very big, output the result modulo $10^9 + 7$

Subtasks & Constraints

For all subtasks, $n, k \geq 0$, and $w_i, s_i \leq 10^9$

Subtask	Points	Additional Constraints
1	3	$n, x \leq 10$
2	7	$n, x \leq 20$
3	10	$n, x \leq 10^3$
4	15	$n \leq 10^6$, $x \leq 2 \cdot 10^5$ and $k = 1$
5	25	$n \leq 10^7$, $x \leq 10^6$ and $k \leq 10^6$
6	40	$n \leq 10^7$, $x \leq 10^6$ and $k \leq 10^7$

k-nacci Sequence

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob has been stuck in his room for days, studying k-nacci sequences. They are defined similarly to the fibonacci sequence. Specifically, a k-nacci sequence of numbers is a sequence of numbers satisfying

$$a_i = b_1 a_{i-1} + b_2 a_{i-2} + \cdots + b_k a_{i-k}$$

You, as Bob's best friend think that it is unhealthy for him to be isolated in his room for so long, so you secretly peek through the dog flap on his door, and see that one main time waster is calculating extra terms of these sequences.

You are given n integers, the first n numbers in the k-nacci sequence, and you need to find xth term of this k-nacci sequence. Note that k is not given, and in your output you can output any valid sequence of b, along with the xth term of this sequence.

Input Format

The first line of input contains n and x, and the next line of input contains $a_1, a_2, \ldots, a_n$, the first n numbers of the given k-nacci sequence.

Output Format

In your first line of output, output a valid sequence of b that matches the integers of the k-nacci sequence. Your next line of output should be the xth term of this sequence. Since the resulting quantity might be big, output it modulo $10^9 + 7$

Subtasks & Constraints

For all subtasks, $n, x, a_i \geq 0$, and $a_i \leq 10^9$.

Subtask	Points	Additional Constraints
1	15	$n, x \leq 10$
2	30	$n, x \leq 10^3$
3	55	$n \leq 10^3, x \leq 10^5$

Making Bank II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Sadly, even after making the optimal amount of money in your career of painting, you are still in debt and do not have any money. As a last result, you have turned your face to *chance games*. One chance game that has gotten you particularly hooked is a modified variant of snakes and ladders, where there are so many snakes, ladders and dice faces that the organisers just represent the game board as a graph. To simplify visualisation of this game, the probabilities are represented as a grid A with n rows and n column. $A_{i,j}$ represents the probability that when you are at square i, you transfer to square j. It is guaranteed that the total probability $A_{i,1} + A_{i,2} + \ldots A_{i,n}$ always equals 1.

Your task is to find the expected number of moves it takes to get from square 1 to n. Since this result might be a fraction, or a big number, you should output 2 integers a and b, the numerator and denominator of the expected number of moves. Also, output a and b modulo $10^9 + 7$.

Input Format

The first line of input contains the integer n. The next n lines of input each contain n space separated integers, together representing A

Output Format

Output the two integers a and b, modulo $10^9 + 7$. If it is not ever possible to get to the last square, output a single integer '-1'

Subtasks & Constraints

For all test cases, $n \geq 1$, $A_{i,j}$ contains at most 10 decimal points, is always between 0 and 1.

Subtask	Points	Additional Constraints
1	3	$n \leq 10$
2	7	$n \leq 40$
3	30	$n \leq 100$
4	60	$n \leq 500$

Sieves

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob likes prime sieves, and knows about lots of them. However, he has recently been getting into some more advanced sieves, like the ones used by his favourite mathematicians in their proofs. He has absolutely no idea how to understand these, so he has now tried understanding the *simple* Min25 sieve! Please implement for Bob, a simple evaluation of the Min25 sieve, using this problem.

$f : \mathbb{N} \to \mathbb{N}$ is a multiplicative function, with $f(p) = p \bmod 6$ for all prime p. Among the values of $f(a), f(a+1), \ldots, f(b)$, find the sum of the subarray within the sequence with the maximal sum.

The input is just a and b, and you should output the subarray with the maximal sum, modulo $10^9 + 7$.

In 10% of the given test cases, $1 \leq a, b \leq 10^1 1$, and for the rest, $0 \leq a, b, \leq 10^1 2$

Formal Power Series Operations

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

This is just practice so you really know how to implement common stuff for FPS's. Basically, you are given the coefficients a and b of 2 polynomials $P(x) = a_0 + a_1x^1 + a_2x^2 + \dots$ and $Q(x) = b_0 + b_1x^1 + b_2x^2 + \dots$. Then, calculate the following:

$$[x^k] \ln \int \sqrt{P(Q(x)^7 7)} \times P(x)^{-1}$$

Input Format

You will first be given the integer $n \leq 10^4$ and the integer $k \leq 10^5$, the number of coefficients in each polynomial, and k as given above. Then the next 2 lines of input contain a and b respectively.

Output Format

Output the above mathematical expression, modulo $10^9 + 7$

Additioning (Troll)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are solving the 'Addition' problem on ORAC, but realise that it is a little different...

You are given an integer N, and you have to find 2 integers A and B which sum up to N. However, you cannot just output *any* pair of integers A and B that sum to N, otherwise this problem would be too easy. Your real task is to find the exact same A and B as Bob, who was solving the addition problem. The only extra bit of information you are given is that $A, B \geq 0$, and that $A \neq B$

Input Format

Your input will just be N

Output Format

Output the two integers A and B

Subtasks & Constraints

Since this is a moderately hard task, for each subtask, you will be given 1% of the available marks if $A+B = N$, an additional 2% of the available marks if your A and B are swapped the wrong way around.

Subtask	Points	Additional Constraints
1	5	$N = 1$
2	7	$N = 3$
3	8	$N = 7$
4	25	$1 \leq N \leq 7$
5	45	$1 \leq N \leq 20$
6	9	$1 \leq N \leq 40$
7	0.9	$1 \leq N \leq 200$
8	0.1	$1 \leq N \leq 2000$

Mixed-up Fractions (Troll)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are studying basic math in trolllang, but keep getting mixed up between different fractions. It is guaranteed that in trolllang that distinct fractions will have distinct representations, and that each representation is a string of unicode characters.

Input Format

You will be given S, the representation of a fraction. Output 'a b', meaning S represents $\frac{a}{b}$. The inventors of trolllang have not discovered negative numbers, so don't worry about them.

Subtasks & Constraints

For each subtask, you will get 1% of the available score if your 'a' is correct, and 3% if your 'b' is correct. Also, each test case is it's own is scored individually in each subtask, weighted equally.

Subtask	Points	Additional Constraints
1	10	$\max(a, b) \leq 1$
2	36	$\max(a, b) \leq 5$
3	54	$\max(a, b) \leq 10$

Sitting xor Standing! (Troll)

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

A local musician is putting on a concert to raise money for charity. The concert will be held in the town hall, a spacious venue perfectly suited for such an event. All of your friends are attending the concert, and at any time, they each have a 98.75% chance of sitting down, and 1.25% chance of not sitting down.

Input Format

You are given a single integer s, your friend's name. You should output either 'Yes' or 'No', to see if that friend is sitting down or not.

Subtasks & Constraints

Each test-case is guaranteed to be a different friend, and each test case is it's own subtask, weighted equally. It is guaranteed that your friend's names' lengths are less than 10^6 letters.